InfoTALAS: Information-Preserving Layer Alignment for Efficient Text Embedding Distillation

Method Proposal for an AAAI-Style Submission

Draft proposal prepared from TALAS and recent 2025-2026 KD literature

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Abstract

Large text embedding models are increasingly used in retrieval-augmented generation, semantic search, and downstream transfer, but their deployment cost motivates distillation into compact encoders. Recent work TALAS distills large embedding teachers into smaller students using teacher-anchored upper-layer alignment, layer-aligned self-distillation, and sharpness-aware optimization. Its key weakness is also its opportunity: layer-aligned self-distillation preserves raw pairwise geometry between adjacent student layers, implicitly treating all batch-wise relations as equally reliable and equally transferable. We propose **InfoTALAS**, a clean extension that does not add multiple extra losses. Instead, it replaces TALAS's raw geometric self-distillation with a single **Information-Preserving Self-Distillation** (IPSD) objective. IPSD converts each layer's batch geometry into semantic neighborhood distributions, aligns adjacent layers via Jensen-Shannon divergence, and gates each anchor by teacher-neighborhood entropy. The resulting method preserves reliable semantic information, attenuates uncertain teacher relations, and imposes an implicit layer-wise information budget through depth-dependent temperatures. The method keeps TALAS's efficiency advantages: it uses only cached teacher sentence embeddings, requires no teacher hidden states, and performs no online teacher inference during student training.

1 One-Sentence Idea

TALAS preserves geometry; InfoTALAS preserves reliable semantic information. Instead of forcing a lower student layer to copy the raw relation matrix of the layer above it, InfoTALAS asks: *which relational information is reliable enough to be transferred, and how much of it should a lower-capacity layer preserve?*

2 Why This Is a Plausible AAAI Motivation

2.1 Starting point: TALAS

TALAS is a strong and practical prior for embedding model distillation. It uses cached teacher sentence embeddings, aligns the upper layers of the student to the teacher embedding through Teacher-Anchored Multi-Layer Distillation (TAMD), propagates knowledge downward through Layer-Aligned Self-Distillation (LASD), and optimizes with ASAM for flatter minima and better generalization [1]. This design avoids expensive online teacher inference and avoids teacher hidden-state storage.

However, the self-distillation component is still geometry-level. For each layer l, TALAS builds a cosine similarity matrix R_l over a mini-batch and minimizes a Frobenius discrepancy between adjacent relation matrices:

$$L_{\text{LASD}} \propto \sum_{l=1}^{L-1} \|R_{l+1} - R_l\|_F^2. \tag{1}$$

This treats every pairwise relation in a batch as a real-valued target of equal status. Under large teacher-student capacity gaps, this can over-transfer noisy neighborhoods, batch artifacts, and over-confident relations into lower student layers.

2.2 Trend in 2025-2026 A*/top-tier KD papers

Recent distillation papers increasingly move from point-wise matching to structure-aware, distribution-aware, and uncertainty-aware transfer. Table 1 summarizes the signals most relevant to the proposed method.

2.3 Gap statement

Existing embedding distillation methods align final embeddings, hidden states, attention patterns, or raw relation matrices. Yet they lack a simple mechanism to quantify **the reliability and amount of relational information** transferred across student layers. This is especially fragile when: (i) the teacher is much larger than the student; (ii) teacher and student have different tokenizers or capacities; (iii) lower student layers should remain flexible and should not inherit overly sharp semantic neighborhoods.

3 Proposed Method: InfoTALAS

3.1 Design principle

The method should be clean enough to defend. We therefore do **not** add three or four new regularizers. We retain the TALAS training skeleton and replace only its self-distillation component:

$$L_{\text{total}} = \lambda_1 L_{\text{SimCSE}} + \lambda_2 L_{\text{TAMD}} + \lambda_3 L_{\text{IPSD}}. \tag{2}$$

ASAM can be kept as the optimizer, following TALAS, but it is not presented as the novelty.

3.2 Intuition

For an anchor sentence x_i in a mini-batch, a layer does not merely produce a vector. It induces a **semantic neighborhood distribution**: which other sentences in the batch are likely neighbors of x_i? If the upper layer believes x_i is close to x_j and far from x_k, the lower layer should preserve that information. But it should not copy all similarities as raw numbers. It should preserve a distributional view of the semantic neighborhood, with reliability controlled by the teacher's uncertainty.

This yields three intuitions:

1. **Geometry to information.** Convert cosine relations into probability distributions over neighbors.
2. **Uniform relations are not useful.** If the teacher neighborhood is nearly uniform, the teacher is uncertain; distillation should be weak for that anchor.
3. **Lower layers need softer information.** A shallow layer should not be forced to maintain the same sharp semantic neighborhood as a deeper layer. A temperature schedule provides an implicit information budget.

3.3 Notation and setup

Let T be a frozen teacher and S be a student with L layers. For input x_i, the cached teacher sentence embedding is $e_i^T \in \mathbb{R}^{d_T}$, and the pooled student embedding at layer l is $e_i^l \in \mathbb{R}^{d_S}$. A mini-batch contains N sentences. We exclude the diagonal self-match when building neighborhood distributions.

3.4 Semantic neighborhood distribution

For each student layer l, anchor i, and neighbor $j \neq i$, define:

$$P_l^{\tau_l}(j \mid i) = \frac{\exp(\cos(e_i^l, e_j^l)/\tau_l)}{\sum_{m \neq i} \exp(\cos(e_i^l, e_m^l)/\tau_l)}. \tag{3}$$

The teacher neighborhood is computed once per batch from cached teacher embeddings:

$$P_T^{\tau_T}(j \mid i) = \frac{\exp(\cos(e_i^T, e_j^T)/\tau_T)}{\sum_{m \neq i} \exp(\cos(e_i^T, e_m^T)/\tau_T)}. \tag{4}$$

This step keeps the offline-teacher benefit of TALAS: it uses only cached teacher sentence embeddings.

3.5 Teacher-uncertainty confidence

We compute the normalized entropy of the teacher neighborhood:

$$\bar{H}_T(i) = \frac{-\sum_{j \neq i} P_T(j \mid i) \log P_T(j \mid i)}{\log(N-1)} \in [0, 1]. \tag{5}$$

Then the confidence gate is:

$$c_i = 1 - \bar{H}_T(i). \tag{6}$$

A low-entropy teacher neighborhood means the teacher provides a specific semantic signal, so c_i is large. A high-entropy neighborhood means the teacher does not strongly distinguish neighbors, so relation transfer is attenuated.

3.6 Implicit layer-wise information budget

We avoid adding an extra entropy loss. Instead, we control the amount of information a layer receives through a depth-dependent temperature:

$$\tau_l = \tau_0 \left(1 + \alpha \frac{L-l}{L-1}\right), \quad \alpha \geq 0. \tag{7}$$

Lower layers have larger τ_l, hence softer and higher-entropy distributions. Upper layers have smaller τ_l, hence sharper semantic neighborhoods closer to the teacher-anchored semantic space.

3.7 Information-Preserving Self-Distillation

We replace TALAS's Frobenius matrix matching with a row-wise distribution matching objective:

$$L_{\text{IPSD}} = \frac{1}{L-1} \sum_{l=1}^{L-1} \frac{1}{N} \sum_{i=1}^{N} c_i \cdot D_{\text{JS}} \left(\text{sg}\left[P_{l+1}^{\tau_{l+1}}(\cdot \mid i)\right], P_l^{\tau_l}(\cdot \mid i)\right), \tag{8}$$

where $\text{sg}[\cdot]$ denotes stop-gradient. The deeper layer acts as the internal teacher, while the lower layer learns a softened, reliability-weighted semantic neighborhood.

The Jensen-Shannon divergence is:

$$D_{\text{JS}}(P, Q) = \frac{1}{2} D_{\text{KL}}(P \| M) + \frac{1}{2} D_{\text{KL}}(Q \| M), \quad M = \frac{1}{2}(P + Q). \tag{9}$$

JS is symmetric, bounded, and more stable than one-way KL when some neighbor probabilities are small.

3.8 What remains from TALAS

The other components remain intentionally close to TALAS:

- **SimCSE regularization:** preserves uniformity and reduces embedding collapse.
- **Teacher-anchored upper-layer distillation:** aligns selected upper student layers to the teacher sentence embedding.
- **ASAM optimization:** can be used to seek flatter minima, as in TALAS.

The proposal is therefore a focused replacement of the self-distillation module rather than an accumulation of losses.

4 Theory and Rationale

4.1 Information view of relation matching

TALAS minimizes squared error between raw relation matrices. InfoTALAS instead views each row of a relation matrix as a conditional distribution over semantic neighbors. This converts layer alignment into preservation of conditional semantic information:

$$x_i \mapsto P_l(\cdot \mid i). \tag{10}$$

The objective is no longer to match every similarity value. It is to preserve the neighbor distribution that determines which examples are semantically close under that layer.

Definition 1 (Semantic neighborhood information)**.** *For an anchor x_i, define its layer-wise semantic neighborhood information as the conditional distribution $P_l(\cdot \mid i)$ over all other batch examples. Preserving this information means keeping the induced distribution close under a divergence such as JS.*

4.2 Why temperature is an information budget

For fixed similarity scores $s_j = \cos(e_i^l, e_j^l)$ and temperature τ, define $P_\tau(j) = \exp(s_j/\tau)/Z_\tau$.

Proposition 1 (Temperature controls entropy)**.** *For fixed scores s, the entropy $H(P_\tau)$ is non-decreasing in τ, with*

$$\frac{d}{d\tau} H(P_\tau) = \frac{\mathrm{Var}_{P_\tau}(s)}{\tau^3} \geq 0. \tag{11}$$

Implication. Increasing τ_l for lower layers gives them softer, higher-entropy targets. Thus Eq. (7) implements an information budget without adding a separate entropy regularizer. Lower layers preserve coarse relational structure; upper layers preserve sharper semantic neighborhoods.

4.3 Why teacher entropy is a reliability signal

If $P_T(\cdot \mid i)$ is uniform, then $H(P_T) = \log(N-1)$ and $c_i = 0$. The teacher provides no reliable neighborhood preference for anchor i, so relation transfer is suppressed. If P_T is concentrated, $H(P_T)$ is low and c_i is high, so the relation signal is trusted.

This is aligned with the general information-theoretic intuition used by entropy-aware KD: samples and relations have different learning values. InfoTALAS applies that idea not to class logits, but to sentence-level semantic neighborhoods.

4.4 Why JS alignment is a stronger target than Frobenius matching

Raw Frobenius matching penalizes absolute similarity differences. In contrast, JS alignment penalizes divergence between normalized neighborhood distributions. This matters because retrieval and sentence embedding tasks are mostly determined by relative neighborhood structure rather than the absolute scale of pairwise similarities.

For any bounded relational query f with $\|f\|_\infty \leq 1$, the difference between expected query values under two neighborhoods is controlled by total variation:

$$|\mathbb{E}_P[f] - \mathbb{E}_Q[f]| \leq 2\,\mathrm{TV}(P, Q). \tag{12}$$

Because JS divergence metrizes distributional closeness and admits Pinsker-type bounds with total variation, minimizing JS controls the discrepancy in downstream neighborhood queries such as top-neighbor preference, local cluster membership, and pairwise retrieval relevance.

4.5 Capacity-gap rationale

A compact student cannot preserve every fine-grained relational detail of a large teacher. TALAS already recognizes this by anchoring only upper layers to the teacher. InfoTALAS extends the same principle inside self-distillation: lower layers receive softer, confidence-weighted relational targets. This prevents the shallow layers from inheriting overly sharp or uncertain neighborhoods while still enabling top-down semantic propagation.

5 Algorithm

Input: Unlabeled corpus D, teacher T, student S, layer count L, upper teacher-anchor depth k, temperatures τ_T, τ_0, schedule parameter α.

1. Precompute and cache teacher sentence embeddings $e_i^T = T(x_i)$ for all $x_i \in D$.
2. For each mini-batch $B = \{x_i\}_{i=1}^N$:
   (a) Retrieve cached teacher embeddings and compute $P_T(\cdot \mid i)$ for each anchor using Eq. (4).
   (b) Compute teacher confidence c_i using Eq. (6).
   (c) Run the student and collect pooled embeddings e_i^l from each layer.
   (d) Compute L_{SimCSE} and L_{TAMD} as in TALAS.
   (e) For each adjacent pair $(l, l+1)$, compute $P_l^{\tau_l}$ and $P_{l+1}^{\tau_{l+1}}$ using Eq. (3).
   (f) Compute L_{IPSD} by Eq. (8).
   (g) Update the student using Eq. (2); optionally use ASAM for the ascent-descent update.

6 Expected Contributions

1. **A clean information-theoretic reinterpretation of layer-wise self-distillation for text embeddings.** We treat each layer's relation structure as semantic neighborhood information rather than a raw similarity matrix.
2. **Information-Preserving Self-Distillation.** IPSD replaces Frobenius matrix matching with JS distribution alignment between adjacent layers.
3. **Teacher-uncertainty-aware relational transfer.** Teacher-neighborhood entropy decides how strongly each anchor's relational information is transferred.
4. **Implicit layer-wise information budgeting.** Depth-dependent temperatures make lower-layer targets softer without introducing additional loss terms.
5. **Practical efficiency.** The method preserves TALAS's offline-teacher setup: no teacher hidden states and no teacher inference during student training.

7 Experimental Plan

7.1 Main comparisons

Use the same teacher-student settings as TALAS where possible:

- Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2 H384.
- BGE-M3 $\rightarrow$ MiniLMv2 H768.
- Qwen3-Embedding-4B $\rightarrow$ BERT-base.

Evaluate on classification, pair classification, and STS benchmarks, with in-domain and out-of-domain averages as in TALAS. If compute allows, add MTEB retrieval or BEIR-style retrieval subsets, because an embedding-distillation paper becomes stronger when retrieval is included.

7.2 Baselines

- Student base and SimCSE-unsupervised.
- TALAS original.
- TALAS with LASD replaced by KL relational matching.
- TALAS with LASD replaced by JS relational matching without confidence.
- Full InfoTALAS: JS + teacher entropy confidence + layer temperature schedule.
- Optional: EMO, DistillCSE, Jasper/Stella, CDM, DSKD if implementation budget allows.

7.3 Ablations

7.4 Diagnostics beyond accuracy

- **Layer entropy curves:** verify lower layers have softer neighborhoods and upper layers become sharper.
- **Teacher-confidence buckets:** evaluate whether low-entropy teacher anchors receive more useful transfer.
- **Out-of-domain gains:** the main claim should be stronger generalization under capacity gap.
- **Memory and time:** show similar memory to TALAS and modest overhead from row-wise JS.
- **Sharpness:** optionally repeat TALAS's Hessian largest-eigenvalue analysis if ASAM is used.

8 Positioning Against TALAS

9 Potential Reviewer Questions and Responses

Q1: Is this just replacing MSE with JS? No. The key change is not the divergence alone. The method redefines layer alignment from raw geometry matching to information-preserving neighborhood transfer, with teacher-entropy reliability and depth-dependent information budgeting. The ablation "TALAS + JS-LASD" should isolate this point.

Q2: Why not maximize mutual information directly? Direct MI estimators such as MINE would add auxiliary networks, instability, and computational overhead. That would conflict with the efficient offline-teacher motivation. IPSD is an information-theoretic surrogate: it preserves conditional neighborhood distributions while staying lightweight.

Q3: Does teacher entropy always measure correctness? No. Entropy measures confidence, not correctness. The claim is more cautious: high entropy means the teacher's neighborhood is uninformative for that anchor, so strong transfer is less justified. Correctness should be evaluated empirically through confidence-bucket analysis.

Q4: Does the method require labels? No. Like TALAS, it can operate in an unsupervised setting using only text inputs and cached teacher sentence embeddings.

Q5: What is the main expected empirical effect? The strongest expected gains should appear in out-of-domain averages and retrieval-like tasks, where preserving reliable semantic neighborhood information matters more than matching raw similarity values.

10 Implementation Notes

- Mask the diagonal $j = i$ before applying softmax.
- Use row-wise JS with small numerical smoothing, e.g. $\epsilon = 10^{-8}$.
- Stop-gradient on P_{l+1} is recommended to prevent both adjacent layers from moving toward a degenerate compromise.
- Start with $\tau_T = 0.05$ or 0.07, $\tau_0 = 0.05$, and $\alpha \in \{0, 0.5, 1.0, 2.0\}$.
- Keep the TALAS loss weights initially, then tune only λ_3 and α.
- Compute P_T from cached teacher embeddings in the same mini-batch. This introduces no online teacher inference.

11 Draft Abstract for a Future Paper

Existing embedding distillation methods typically transfer point-wise teacher embeddings or preserve raw relational geometry across layers. However, under large teacher-student capacity gaps, not all relational signals are equally reliable or equally transferable. We propose InfoTALAS, an information-preserving layer alignment framework for efficient text embedding distillation. Instead of forcing adjacent student layers to match raw similarity matrices, InfoTALAS models each layer's batch geometry as a semantic neighborhood distribution and transfers it through a teacher-uncertainty-aware Jensen-Shannon objective. A depth-dependent temperature schedule further imposes an implicit information budget, allowing lower layers to preserve coarse semantic structure while upper layers preserve sharper teacher-aligned neighborhoods. InfoTALAS requires only cached teacher sentence embeddings and avoids teacher hidden states or online teacher inference. Experiments across sentence embedding benchmarks are expected to show improved out-of-domain generalization with minimal additional memory overhead.

12 Milestone Plan

1. **Week 1:** Implement IPSD module and validate numerical stability on one TALAS teacher-student pair.
2. **Week 2:** Run core ablations: Frobenius vs JS, with/without confidence, constant vs depth temperature.
3. **Week 3:** Run all TALAS benchmark datasets; produce Avg-In, Avg-Out, Avg-All tables.
4. **Week 4:** Add diagnostics: entropy curves, teacher-confidence buckets, memory/time, optional sharpness analysis.
5. **Week 5:** Add retrieval/MTEB subset if compute allows; finalize related work and method theory.

13 Summary

InfoTALAS is a focused, defensible extension of TALAS. It keeps the efficient offline-teacher training pipeline but replaces geometry-level layer-aligned self-distillation with a single information-preserving objective. The central claim is simple: in embedding distillation, lower layers should not copy all raw relations from upper layers; they should preserve reliable semantic neighborhood information under a capacity-aware information budget.

References

[1] Quoc Phong Dao, Hoang Son Nguyen, Pham Khanh Chi, Linh Ngo Van, Diep Thi-Ngoc Nguyen, Thien Huu Nguyen, and Trung Le. 2026. TALAS: Teacher-Anchored Layer Alignment with Adaptive Sharpness-Aware Minimization for Embedding Distillation. Uploaded paper analyzed in this proposal.

[2] Minh-Phuc Truong, Hai An Vu, Tu Vu, Nguyen Thi Ngoc Diep, Linh Ngo Van, Thien Huu Nguyen, and Trung Le. 2025. EMO: Embedding Model Distillation via Intra-Model Relation and Optimal Transport Alignments. In *Proceedings of EMNLP 2025*, pages 7594–7606.

[3] Xiao Cui, Mo Zhu, Yulei Qin, Liang Xie, Wengang Zhou, and Houqiang Li. 2025. Multi-Level Optimal Transport for Universal Cross-Tokenizer Knowledge Distillation on Language Models. In *Proceedings of AAAI 2025*. Also arXiv:2412.14528.

[4] Hoang Tran Vuong, Tue Le, Quyen Tran, Linh Ngo Van, and Trung Le. 2026. MCW-KD: Multi-Cost Wasserstein Knowledge Distillation for Large Language Models. In *Proceedings of AAAI 2026*, 40(39):33332–33340.

[5] Hengzhi Wang, Yucheng Jing, Hongsong Sun, Hao Wang, Jiahao Liao, Leszek Rutkowski, and Dacheng Tao. 2026. Bridging the Tokenizer Gap: Semantics and Distribution-aware Knowledge Transfer for Unbiased Cross-Tokenizer Distillation. In *Proceedings of AAAI 2026*, 40(39):33494–33502.

[6] Yuanhong Zhang, Muyao Yuan, Weizhan Zhang, Tieliang Gong, Wen Wen, Jiangyong Ying, and Weijie Shi. 2025. InfoSAM: Fine-Tuning the Segment Anything Model from an Information-Theoretic Perspective. *ICML 2025 Spotlight*.

[7] Chi-Ping Su, Ching-Hsun Tseng, Bin Pu, Lei Zhao, Jiewen Yang, Zhuangzhuang Chen, and Shin-Jye Lee. 2025. EA-KD: Entropy-based Adaptive Knowledge Distillation. In *Proceedings of ICCV 2025*.

[8] Shuya Feng and Yuan Hong. 2025. DPED: Multi-Layer Noise Distillation for Privacy-Preserving Text Embeddings. In *Proceedings of EMNLP 2025*, pages 25237–25245.

[9] Robin Vujanic and Thomas Rueckstiess. 2025. LEAF: Knowledge Distillation of Text Embedding Models with Teacher-Aligned Representations. arXiv:2509.12539.

[10] Tianyu Gao, Xingcheng Yao, and Danqi Chen. 2022. SimCSE: Simple Contrastive Learning of Sentence Embeddings. In *EMNLP 2021*.

[11] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the Knowledge in a Neural Network. arXiv:1503.02531.

[12] Jungmin Kwon, Jeongseop Kim, Hyunseo Park, and In Kwon Choi. 2021. ASAM: Adaptive Sharpness-Aware Minimization for Scale-Invariant Learning of Deep Neural Networks. In *ICML 2021*.

Table 1: Recent signals motivating information-theoretic self-distillation. The main trend is not simply to match vectors, but to control distributional structure, relation reliability, and transfer granularity.

Paper	**Core signal**	**Implication for InfoTALAS**
TALAS 2026 [1]	Cached teacher embeddings, upper-layer teacher anchoring, relation-based self-distillation, ASAM.	Efficient embedding distillation can avoid online teacher inference; the open gap is how to control relation transfer across student layers.
EMO, EMNLP 2025 [2]	Cross-tokenizer embedding distillation with intra-model relation alignment and optimal transport over important hidden states.	Embedding KD benefits from structural and semantic importance signals, not only output imitation.
MultiLevelOT, AAAI 2025 [3]	Aligns teacher-student logit distributions at token and sequence levels using OT and Sinkhorn distance.	Cross-tokenizer KD increasingly models distributional geometry rather than one-to-one matching.
MCW-KD, AAAI 2026 [4]	Uses multiple Wasserstein cost functions to align hidden states and output distributions under heterogeneity.	A single raw distance can be insufficient when model discrepancies are multifaceted.
SeDi, AAAI 2026 [5]	Explicitly targets semantic information, biased logit alignment, and distributional structure; aligns student entropy with teacher entropy.	Entropy is a credible information-theoretic handle for controlling distillation reliability.
InfoSAM, ICML 2025 Spotlight [6]	Uses mutual-information-based objectives to preserve domain-invariant relations during foundation-model adaptation.	Information theory can justify preserving useful relations while filtering pseudo-invariant or irrelevant information.
EA-KD, ICCV 2025 [7]	Uses entropy to adaptively reweight KD samples because samples have different learning values.	A teacher relation should not be trusted uniformly; entropy can quantify reliability.
DPED, EMNLP 2025 [8]	Multi-layer noise distillation for private text embeddings.	Embedding distillation must care about which information is preserved, injected, or suppressed.

Table 2: Minimal ablation plan designed to avoid the appearance of loss engineering.

Variant	**Question answered**
TALAS	How strong is the direct prior baseline?
TALAS + JS-LASD	Is distribution matching alone better than Frobenius geometry matching?
IPSD without c_i	Does teacher-entropy confidence matter?
IPSD with constant τ	Does the layer-wise information budget matter?
IPSD without stop-gradient	Does using a detached upper-layer guide stabilize learning?
Full InfoTALAS	Do all IPSD components combine effectively?

Table 3: Core distinction from TALAS.

Aspect	**TALAS**	**InfoTALAS**
Self-distillation target	Raw cosine relation matrix.	Semantic neighborhood distribution.
Distance	Frobenius/MSE over matrix entries.	JS divergence over conditional neighbor distributions.
Reliability control	All anchors and relations weighted uniformly.	Teacher-neighborhood entropy gates unreliable anchors.
Layer capacity control	Same relation-matching principle across depth.	Lower layers receive softer targets through larger temperatures.
Loss count	SimCSE + TAMD + LASD.	SimCSE + TAMD + IPSD. No extra loss count.
Efficiency	Offline teacher embeddings.	Same offline-teacher setup.